

VECTOR 3D

SINGLE CORRECT TYPE QUESTIONS

1. The vector $\vec{a} = -\hat{i} + 2\hat{j} + \hat{k}$ is rotated through a right angle, passing through the y-axis in its way and the resulting vector is $\vec{b}$. Then the projection of $3\vec{a} + \sqrt{2}\vec{b}$ on $\vec{c} = 5\hat{i} + 4\hat{j} + 3\hat{k}$ is :

सदिश $\vec{a} = -\hat{i} + 2\hat{j} + \hat{k}$ को y-अक्ष से होकर ले जाते हुए एक समकोण तक घुमाया जाता है और इसके परिणामस्वरूप सदिश $\vec{b}$ प्राप्त होता है। तो $3\vec{a} + \sqrt{2}\vec{b}$ का सदिश $\vec{c} = 5\hat{i} + 4\hat{j} + 3\hat{k}$ पर प्रक्षेप है :

- (A) $\sqrt{6}$ (B) $2\sqrt{3}$ (C) 1 (D) $3\sqrt{2}$

Ans. D

Sol. $\vec{b} = \lambda \vec{a} \times (\vec{a} \times \hat{j})$

$$\Rightarrow \vec{b} = \lambda(-2\hat{i} - 2\hat{j} + 2\hat{k})$$

$$|\vec{b}| = |\vec{a}| \quad \therefore \sqrt{6} = \sqrt{12} |\lambda| \Rightarrow \lambda = \pm \frac{1}{\sqrt{2}}$$

$$\left(\lambda = \frac{1}{\sqrt{2}} \text{ rejected } \because \vec{b} \text{ makes acute angle with y axis} \right)$$

$$\vec{b} = -\sqrt{2}(-\hat{i} - \hat{j} + \hat{k})$$

$$\frac{(3\vec{a} + \sqrt{2}\vec{b}) \cdot \vec{c}}{|\vec{c}|} = 3\sqrt{2}$$

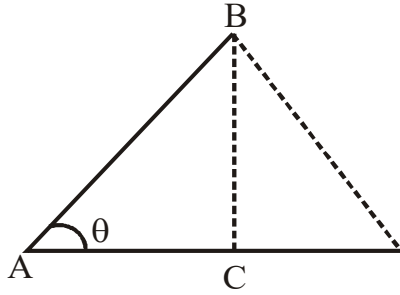
2. In a triangle ABC. If $|\overline{BC}| = 8, |\overline{CA}| = 7, |\overline{AB}| = 10$, then the projection of the vector $\overline{AB}$ on $\overline{AC}$ is equal to:

एक त्रिभुज ABC में यदि $|\overline{BC}| = 8, |\overline{CA}| = 7, |\overline{AB}| = 10$ है, तो सदिश $\overline{AB}$ का सदिश $\overline{AC}$ पर प्रक्षेप बराबर है :

- (A) $\frac{127}{20}$ (B) $\frac{115}{16}$ (C) $\frac{25}{4}$ (D) $\frac{85}{14}$

Ans. D

Sol.



$$|\vec{BC}| = 8 \quad |\vec{CA}| = 7 \quad |\vec{AB}| = 10$$

$$\text{Projection} = |\vec{AB}| \cos \theta$$

$$\text{Projection} = 10 \cos \theta$$

$$\cos \theta = \frac{|\vec{AB}|^2 + |\vec{CA}|^2 - |\vec{BC}|^2}{2 \times |\vec{AB}| |\vec{CA}|}$$

$$= \frac{100 + 49 - 64}{2 \times 10 \times 7} = \frac{85}{10 \times 2 \times 7}$$

$$\text{Projection} = 10 \times \frac{85}{10 \times 2 \times 7} = \frac{85}{14}$$

3. If (x, y, z) be an arbitrary point lying on a plane P which passes through the points $(42, 0, 0)$, $(0, 42, 0)$ and $(0, 0, 42)$, then the value of the expression

$$3 + \frac{x-11}{(y-19)^2(z-12)^2} + \frac{y-19}{(x-11)^2(z-12)^2} + \frac{z-12}{(x-11)^2(y-19)^2} - \frac{x+y+z}{14(x-11)(y-19)(z-12)} \text{ is equal to :}$$

माना बिन्दुओं $(42, 0, 0)$, $(0, 42, 0)$ तथा $(0, 0, 42)$ से होकर जाने वाले समतल P पर (x, y, z) एक स्वेच्छ बिन्दु हैं, तो व्यंजक

$$3 + \frac{x-11}{(y-19)^2(z-12)^2} + \frac{y-19}{(x-11)^2(z-12)^2} + \frac{z-12}{(x-11)^2(y-19)^2} - \frac{x+y+z}{14(x-11)(y-19)(z-12)} \text{ का मान है :}$$

- (A) -45 (B) 39 (C) 3 (D) 0

Ans. C

Sol. $P : x + y + z = 42$

$$(x-11) + (y-19) + (z-12) = 0$$

$$\text{Let } x-11 = a$$

$$y-19 = b$$

$$z-12 = c$$

$$a + b + c = 0$$

$$3 + \frac{a}{b^2c^2} + \frac{b}{a^2c^2} + \frac{c}{a^2b^2} - \frac{42}{14abc}$$

$$\Rightarrow 3 + \frac{a^3 + b^3 + c^3}{a^2b^2c^2} - \frac{3}{abc}$$

$$\Rightarrow 3 + \frac{3abc}{a^2b^2c^2} - \frac{3}{abc}$$

$$\Rightarrow 3$$

4. A vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ ($\alpha, \beta \in \mathbb{R}$) lies in the plane of the vectors, $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} + 4\hat{k}$. If $\vec{a}$ bisects the angle between $\vec{b}$ and $\vec{c}$, then :

एक सदिश $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ ($\alpha, \beta \in \mathbb{R}$) उस समतल में, जिसमें दोनों सदिश $\vec{b} = \hat{i} + \hat{j}$ तथा $\vec{c} = \hat{i} - \hat{j} + 4\hat{k}$ स्थित हैं, स्थित है। यदि $\vec{a}$ सदिशों $\vec{b}$ और $\vec{c}$ के बीच के कोण को समद्विभाजित करता है, तो :

- (A) $\vec{a} \cdot \hat{k} + 4 = 0$ (B) $\vec{a} \cdot \hat{k} + 2 = 0$ (C) $\vec{a} \cdot \hat{i} + 1 = 0$ (D) $\vec{a} \cdot \hat{i} + 3 = 0$

Ans. B

Sol. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar $\Rightarrow \begin{vmatrix} \alpha & 1 & \beta \\ 1 & 1 & 0 \\ 1 & -1 & 4 \end{vmatrix} = 0$ (1)

Also $\vec{a}$ bisects the angle between $\vec{b}$ and $\vec{c}$

$$\Rightarrow \text{Let } \vec{a} = \lambda (\vec{b} + \vec{c}) = \lambda \left(\frac{\hat{i} + \hat{j}}{3\sqrt{2}} + \frac{\hat{i} - \hat{j} + 4\hat{k}}{3\sqrt{2}} \right)$$

$$\Rightarrow \alpha\hat{i} + 2\hat{j} + \beta\hat{k} = \lambda \left(\frac{4}{3\sqrt{2}}\hat{i} + \frac{2}{3\sqrt{2}}\hat{j} + \frac{4}{3\sqrt{2}}\hat{k} \right)$$

Comparing, $\frac{2\lambda}{3\sqrt{2}} = 2 \Rightarrow \lambda = 3\sqrt{2}$

$$\alpha = 4; \beta = 4 \quad \vec{a} = 4\hat{i} + 2\hat{j} + 4\hat{k}$$

But it does not satisfy any option

$$\text{So let } \vec{a} = \mu (\vec{b} - \vec{c}) = \mu \left(\frac{2}{3\sqrt{2}}\hat{i} + \frac{4}{3\sqrt{2}}\hat{j} - \frac{4}{3\sqrt{2}}\hat{k} \right)$$

Comparing, $\mu = \frac{3\sqrt{2}}{2} \Rightarrow \vec{a} = \hat{i} + 2\hat{j} - 2\hat{k}$

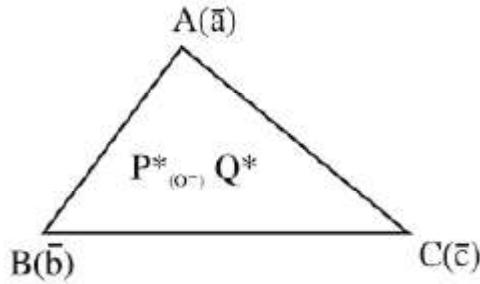
$$\Rightarrow \vec{a} \cdot \hat{k} = -2 \quad \Rightarrow \vec{a} \cdot \hat{k} + 2 = 0$$

5. If the points P and Q are respectively the circumcenter and the orthocentre of a ΔABC , then $\vec{PA} + \vec{PB} + \vec{PC}$ is :

यदि एक त्रिभुज ABC के परिकेन्द्र तथा लंबकेन्द्र क्रमशः P तथा Q है, तो $\vec{PA} + \vec{PB} + \vec{PC}$ बराबर है :

- (A) $2\vec{PQ}$ (B) $\vec{QP}$ (C) $2\vec{QP}$ (D) $\vec{PQ}$

Ans. D



Sol.

$$\vec{PA} + \vec{PB} + \vec{PC} = \vec{a} + \vec{b} + \vec{c}$$

$$\vec{PG} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

$$\Rightarrow \vec{a} + \vec{b} + \vec{c} = 3\vec{PG} = \vec{PQ}$$

6. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non zero vectors such that $\vec{b} \cdot \vec{c} = 0$ and $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b} - \vec{c}}{2}$. If $\vec{d}$ be a vector such that $\vec{b} \cdot \vec{d} = \vec{a} \cdot \vec{b}$, then $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$ is equal to :

माना $\vec{a}$, $\vec{b}$ और $\vec{c}$ तीन शून्येतर सदिश हैं जबकि $\vec{b} \cdot \vec{c} = 0$ और $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b} - \vec{c}}{2}$ । यदि $\vec{d}$ एक सदिश ऐसा है कि $\vec{b} \cdot \vec{d} = \vec{a} \cdot \vec{b}$ तब

$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$ बराबर होगा—

- (A) $\frac{1}{4}$ (B) $\frac{3}{4}$ (C) $-\frac{1}{4}$ (D) $\frac{1}{2}$

Ans. A

Sol. $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b} - \vec{c}}{2} \Rightarrow (\vec{a}, \vec{c})\vec{b} - (\vec{a}, \vec{b})\vec{c} = \frac{\vec{b}}{2} - \frac{\vec{c}}{2}$

$$\Rightarrow \vec{a} \cdot \vec{c} = \frac{1}{2}, \vec{a} \cdot \vec{b} = \frac{1}{2} (\because \vec{b} \cdot \vec{c} = 0)$$

Given $\vec{b} \cdot \vec{d} = \vec{a} \cdot \vec{b}$

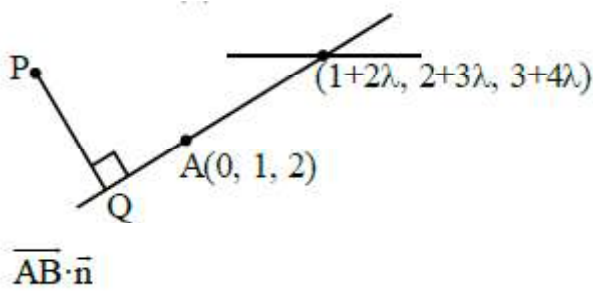
$$\text{Now } (\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c}) = \frac{1}{4}$$

7. Let the line L pass through the point (0,1,2), intersect the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and be parallel to the plane $2x + y - 3z = 4$. Then the distance of the point P(1, -9, 2) from the line L is :

माना रेखा L, बिंदु (0,1,2) से होकर जाती है, रेखा $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ को काटती है तथा समतल $2x + y - 3z = 4$ समांतर है। तो बिंदु P(1, -9, 2) की रेखा L से दूरी है :

- (A) $\sqrt{74}$ (B) 9 (C) $\sqrt{69}$ (D) $\sqrt{54}$

Ans. A



Sol.

$$\Rightarrow [(1+2\lambda)\hat{i} + (1+3\lambda)\hat{j} + (1+4\lambda)\hat{k}] \cdot (2\hat{i} + \hat{j} - 3\hat{k})$$

$$2 + 4\lambda + 1 + 3\lambda - 3 - 12\lambda = 0$$

$$5\lambda = 0 \Rightarrow \lambda = 0$$

$$\text{Line } \vec{AB}, \vec{r} = \hat{j} + 2\hat{k} + \mu(\hat{i} + \hat{j} + \hat{k})$$

$$\text{General form : } Q(\mu, 1+\mu, 2+\mu)$$

$$\therefore \vec{PQ} \cdot \vec{AB} = 0$$

$$(\mu - 1) + (10 + \mu) + \mu = 0$$

$$3\mu = -9 \Rightarrow \mu = -3$$

$$\therefore \text{distance} = \sqrt{16 + 49 + 9} = \sqrt{74}$$

8. Let $\vec{u} = \hat{i} - \hat{j} - 2\hat{k}$, $\vec{v} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{v} \cdot \vec{w} = 2$ and $\vec{v} \times \vec{w} = \vec{u} + \lambda \vec{v}$. Then $\vec{u} \cdot \vec{w}$ is equal to :

माना $\vec{u} = \hat{i} - \hat{j} - 2\hat{k}$, $\vec{v} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{v} \cdot \vec{w} = 2$ तथा $\vec{v} \times \vec{w} = \vec{u} + \lambda \vec{v}$. तो $\vec{u} \cdot \vec{w}$ बराबर है :

- (A) $-\frac{2}{3}$ (B) $\frac{3}{2}$ (C) 2 (D) 1

Ans. D

Sol. $\vec{v} \times \vec{w} = \vec{u} + \lambda \vec{v}$

$$(\vec{v} \times \vec{w}) \cdot \vec{v} = \vec{u} \cdot \vec{v} + \lambda \vec{v} \cdot \vec{v}$$

$$0 = \vec{u} \cdot \vec{v} + \lambda |\vec{v}|^2$$

$$\lambda = -\left(\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2}\right) = -\left(\frac{2-1+2}{6}\right) = -\frac{1}{2}$$

$$\text{Now, } (\vec{v} \times \vec{w}) \cdot \vec{w} = \vec{u} \cdot \vec{w} + \lambda \vec{v} \cdot \vec{w}$$

$$\Rightarrow 0 = \vec{u} \cdot \vec{w} + 2\lambda \Rightarrow \vec{u} \cdot \vec{w} = -2\lambda$$

$$\Rightarrow \vec{u} \cdot \vec{w} = -2\left(-\frac{1}{2}\right) = 1$$

9. The plane $2x - y + z = 4$ intersects the line segment joining the points A ($a, -2, 4$) and B ($2, b, -3$) at the point C in the ratio 2 : 1 and the distance of the point C from the origin is $\sqrt{5}$. If $ab < 0$ and P is the point ($a - b, b, 2b - a$) then CP^2 is equal to

समतल $2x - y + z = 4$ बिंदुओं A ($a, -2, 4$) तथा B ($2, b, -3$) को मिलाने वाले रेखाखंड को 2 : 1 के अनुपात में बिंदु C पर काटता है तथा मूलबिंदु से बिंदु C की दूरी $\sqrt{5}$ है। यदि $ab < 0$ है तथा P, बिंदु ($a - b, b, 2b - a$) है, तो CP^2 बराबर है :

- (A) $\frac{16}{3}$ (B) $\frac{97}{3}$ (C) $\frac{17}{3}$ (D) $\frac{73}{3}$

Ans. C

Sol. A($a, -2, 4$), B($2, b, -3$)

$$AC : CB = 2 : 1$$

$$\Rightarrow C \equiv \left(\frac{a+4}{3}, \frac{2b-2}{3}, \frac{-2}{3}\right)$$

$$C \text{ lies on } 2x - y + z = 4$$

$$\Rightarrow \frac{2a+8}{3} - \frac{2b-2}{3} - \frac{2}{3} = 4$$

$$\Rightarrow a - b = 2 \dots (1)$$

$$\text{Also } OC = \sqrt{5}$$

$$\Rightarrow \left(\frac{a+4}{3}\right)^2 + \left(\frac{2b-2}{3}\right)^2 + \frac{4}{9} = 5 \quad \dots(2)$$

Solving, (A) and (2)

$$(b+6)^2 + (2b-2)^2 = 41$$

$$\Rightarrow 5b^2 + 4b - 1 = 0$$

$$\Rightarrow b = -1 \text{ or } \frac{1}{5}$$

$$\Rightarrow a = 1 \text{ or } \frac{11}{5}$$

$$\text{But } ab < 0 \Rightarrow (a, b) = (1, -1)$$

$$C \equiv \left(\frac{5}{3}, \frac{-4}{3}, \frac{-2}{3}\right), P \equiv (2, -1, -3)$$

$$CP^2 = \frac{1}{9} + \frac{1}{9} + \frac{49}{9} = \frac{51}{9} = \frac{17}{3}$$

10. A vector $\vec{a}$ has components $3p$ and 1 with respect to a rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to new system, $\vec{a}$ has components $p+1$ and $\sqrt{10}$, then a value of p is equal to :

एक आयताकार कार्तीय प्रणाली के सापेक्ष एक सदिश $\vec{a}$ के घटक $3p$ तथा 1 हैं। इस प्रणाली को मूलबिन्दु के सापेक्ष वामावृत्त दिशा में किसी कोण तक घुमाया जाता है। यदि नई प्रणाली के सापेक्ष $\vec{a}$ के घटक $p+1$ तथा $\sqrt{10}$ हैं, तो p का एक मान बराबर है :

- (A) 1 (B) $\frac{4}{5}$ (C) $-\frac{5}{4}$ (D) -1

Ans. D

Sol. Since lengths do not change

$$\Rightarrow \sqrt{(3p)^2 + (1)^2} = \sqrt{(p+1)^2 + (\sqrt{10})^2}$$

$$\Rightarrow 9p^2 + 1 = p^2 + 2p + 1 + 10$$

$$\Rightarrow 8p^2 - 2p - 10 = 0 \Rightarrow 4p^2 - p - 5 = 0$$

$$\Rightarrow 4p^2 - 5p + 4p - 5 = 0$$

$$\Rightarrow p = -1 \text{ or } p = \frac{5}{4}$$

11. Let $a, b, c \in \mathbb{R}$ be such that $a^2 + b^2 + c^2 = 1$. If $a \cos \theta = b \cos \left(\theta + \frac{2\pi}{3} \right) = c \cos \left(\theta + \frac{4\pi}{3} \right)$, where $\theta = \frac{\pi}{9}$, then the angle between the vectors $a\hat{i} + b\hat{j} + c\hat{k}$ and $b\hat{i} + c\hat{j} + a\hat{k}$ is :

माना $a, b, c \in \mathbb{R}$, जिनके लिए $a^2 + b^2 + c^2 = 1$ है। यदि $a \cos \theta = b \cos \left(\theta + \frac{2\pi}{3} \right) = c \cos \left(\theta + \frac{4\pi}{3} \right)$ है, जबकि $\theta = \frac{\pi}{9}$ है, तो सदिशों $a\hat{i} + b\hat{j} + c\hat{k}$ तथा $b\hat{i} + c\hat{j} + a\hat{k}$ के बीच का कोण है :

- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{9}$ (C) $\frac{2\pi}{3}$ (D) 0

Ans. A

Sol. $a \cos \theta = b \cos \left(\theta + \frac{2\pi}{3} \right) = c \cos \left(\theta + \frac{4\pi}{3} \right) = K$

$$\frac{K}{a} = \cos \theta$$

$$\frac{K}{b} = \cos \left(\frac{2\pi}{3} + \theta \right)$$

$$\frac{K}{c} = \cos \left(\frac{4\pi}{3} + \theta \right)$$

$$\frac{K}{a} + \frac{K}{b} + \frac{K}{c} = \cos \theta + \left(\cos \frac{2\pi}{3} + \cos \theta \right) + \cos \left(\frac{4\pi}{3} + \theta \right)$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$$

$$ab + bc + ca = 0$$

$$\cos \alpha = \frac{ab + bc + ca}{\sqrt{a^2 + b^2 + c^2} \sqrt{a^2 + b^2 + c^2}}$$

$$\cos \alpha = 0$$

$$= \frac{\pi}{2}$$

12. Let $\lambda \in \mathbb{R}$, $\vec{a} = \lambda\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{b} = \hat{i} - \lambda\hat{j} + 2\hat{k}$. If $\left((\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b}) \right) \times (\vec{a} - \vec{b}) = 8\hat{i} - 40\hat{j} - 24\hat{k}$,

then $\left| \lambda (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) \right|^2$ is equal to

माना $\lambda \in \mathbb{R}$, $\vec{a} = \lambda\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{b} = \hat{i} - \lambda\hat{j} + 2\hat{k}$ है। यदि $\left((\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b})\right) \times (\vec{a} - \vec{b}) = 8\hat{i} - 40\hat{j} - 24\hat{k}$ है,

तब $\left|\lambda(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})\right|^2$ बराबर है :

(A) 136

(B) 140

(C) 144

(D) 132

Ans. B

Sol. $\vec{a} = \lambda\hat{i} + 2\hat{j} - 3\hat{k}$

$$\vec{b} = \hat{i} - \lambda\hat{j} + 2\hat{k}$$

$$\Rightarrow (\vec{b} - \vec{a}) \times ((\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b})) = 8\hat{i} - 40\hat{j} - 24\hat{k}$$

$$\Rightarrow ((\vec{a} - \vec{b})(\vec{a} + \vec{b}))(\vec{a} \times \vec{b}) = 8\hat{i} - 40\hat{j} - 24\hat{k}$$

$$\Rightarrow 8(\vec{a} \times \vec{b}) = 8\hat{i} - 40\hat{j} - 24\hat{k}$$

$$\text{Now, } \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \lambda & 2 & -3 \\ 1 & -\lambda & 2 \end{vmatrix}$$

$$= (4 - 3\lambda)\hat{i} - (2\lambda + 3)\hat{j} + (-\lambda^2 - 2)\hat{k}$$

$$\Rightarrow \lambda = 1$$

$$\therefore \vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$$

$$\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$$

$$\Rightarrow \vec{a} + \vec{b} = 2\hat{i} + \hat{j} - \hat{k}, \vec{a} - \vec{b} = 3\hat{j} - 5\hat{k}$$

$$\Rightarrow (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 0 & 3 & -5 \end{vmatrix} = 2\hat{i} + 10\hat{j} + 6\hat{k}$$

$$\therefore \text{required answer} = 4 + 100 + 36 = 140$$

13. If the equation of the plane containing the line $x + 2y + 3z - 4 = 0 = 2x + y - z + 5$ and perpendicular to the plane $\vec{r} = (\hat{i} - \hat{j}) + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$ is $ax + by + cz = 4$, then $(a - b + c)$ is equal to :

यदि समतल, जिसमें रेखा $x + 2y + 3z - 4 = 0 = 2x + y - z + 5$ स्थित है तथा जो समतल

$\vec{r} = (\hat{i} - \hat{j}) + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$ के लंबवत् है, का समीकरण $ax + by + cz = 4$ है, तो $(a - b + c)$ बराबर है :

(A) 18

(B) 22

(C) 20

(D) 24

Ans. B

Sol. Equation of plane containing line of intersection of plane $x + 2y + 3z - 4 = 0$ and $2x + y - z + 5 = 0$ will be $P_1 + \lambda P_2 = 0 \Rightarrow (1 + 2\lambda)x + (2 + \lambda)y + (3 - \lambda)z + (5\lambda - 4) = 0$ _____(1)

This plane is $\perp$ to plane

$$\vec{r} = (\vec{i} - \vec{j}) + \lambda(\vec{i} + \vec{j} + \vec{k}) + \mu(\vec{i} - 2\vec{j} + 3\vec{k}) \quad \text{_____}(2)$$

$$\text{Normal of plane (2)} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -2 & 3 \end{vmatrix} = 5\hat{i} - 2\hat{j} - 3\hat{k}$$

Now both plane (1) and (2) $\perp$ Hence their normal also $\perp$

$$5(1 + 2\lambda) - 2(2 + \lambda) - 3(3 - \lambda) = 0$$

$$-8 + 11\lambda = 0 \Rightarrow \lambda = \frac{8}{11}$$

$$\text{Now equation of plane } \frac{27}{11}x + \frac{30}{11}y + \frac{25}{11}z - \frac{4}{11} = 0$$

$$27x + 30y + 25z = 4$$

$$\text{Hence } a - b + c = 27 - 30 + 25 = 22$$

14. If for $a > 0$, the feet of perpendiculars from the points $A(a, -2a, 3)$ and $B(0, 4, 5)$ on the plane $lx + my + nz = 0$ are points $C(0, -a, -1)$ and D respectively, then the length of line segment CD is equal to:

यदि $a > 0$ के लिए, बिन्दुओं $A(a, -2a, 3)$ तथा $B(0, 4, 5)$ से समतल $lx + my + nz = 0$ पर लम्बों के पाद क्रमशः बिन्दु $C(0, -a, -1)$ तथा D हैं, तो रेखा खण्ड CD की लम्बाई है :

(A) $\sqrt{55}$

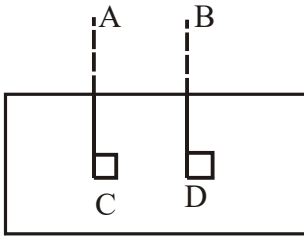
(B) $\sqrt{31}$

(C) $\sqrt{66}$

(D) $\sqrt{41}$

Ans. C

Sol.



$\vec{n}$ = normal vector of plane $\overrightarrow{AC} = (a, -a, -4)$

equation of plane

$$ax - ay + 4z = 0$$

$C(0, -a, -1)$ lies on it

$$0 + a^2 - 4 = 0 \Rightarrow a = 2$$

equation of plane $x - y + 2z = 0$

$$C(0, -2, -1)$$

equation of line BD

$$\vec{r} = (0, 4, 5) + \lambda (1, -1, 2)$$

$$D(\lambda, 4 - \lambda, 5 + 2\lambda)$$

D will lie on Plane $x - y + 2z = 0$

$$\lambda - (4 - \lambda) + 2(5 + 2\lambda) = 0$$

$$6\lambda = -6$$

$$\lambda = -1$$

$$D(-1, 5, 3)$$

$$|\overrightarrow{CD}| = \sqrt{1+49+16} = \sqrt{66}$$

15. Let ABCD be a quadrilateral. If E and F are the mid points of the diagonals AC and BD respectively and $(\overrightarrow{AB} - \overrightarrow{BC}) + (\overrightarrow{AD} - \overrightarrow{DC}) = k \overrightarrow{FE}$, then k is equal to :

माना ABCD एक चतुर्भुज है जिसके विकर्ण AC तथा BD के मध्य बिन्दु क्रमशः E तथा F है, तथा $(\overrightarrow{AB} - \overrightarrow{BC}) + (\overrightarrow{AD} - \overrightarrow{DC}) = k \overrightarrow{FE}$ हो तो k का मान होगा :

(A) 4

(B) -2

(C) 2

(D) -4

Ans. D

Sol. Let P.V. & A, B, C, D are $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ resp. mid-point of AC is E.

$$\Rightarrow \text{P.V. of E are } \left(\frac{\vec{a} + \vec{c}}{2} \right)$$

Mid point of BD is F

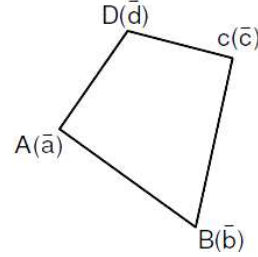
$$\Rightarrow \text{P.V. of F are } \left(\frac{\vec{b} + \vec{d}}{2} \right)$$

$$(\vec{AB} - \vec{BC}) + (\vec{AD} - \vec{DC}) = (\vec{b} - \vec{a}) - (\vec{c} - \vec{b}) + (\vec{d} - \vec{a}) - (\vec{c} - \vec{d})$$

$$= 2\vec{b} - 2\vec{a} - 2\vec{c} + 2\vec{d}$$

$$= 2(\vec{b} + \vec{d}) - 2(\vec{a} + \vec{c})$$

$$= 4 \left[\left(\frac{\vec{b} + \vec{d}}{2} \right) - \left(\frac{\vec{a} + \vec{c}}{2} \right) \right] = 4\vec{EF} = -4\vec{FE}$$



16. Let P be the point of intersection of the line $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2}$ and the plane $x + y + z = 2$. If the distance of the point P from the plane $3x - 4y + 12z = 32$ is q, then q and 2q are the roots of the equation:

माना रेखा $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2}$ तथा समतल $x + y + z = 2$ का प्रतिच्छेदन बिंदु P है। यदि बिंदु P की समतल $3x - 4y + 12z = 32$

से दूरी q है, तो q तथा 2q किस समीकरण के मूल हैं ?

- (A) $x^2 - 18x - 72 = 0$ (B) $x^2 + 18x - 72 = 0$ (C) $x^2 + 18x + 72 = 0$ (D) $x^2 - 18x + 72 = 0$

Ans. D

Sol. $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2} = \lambda$

$$A(3\lambda - 3, \lambda - 2, 1 - 2\lambda)$$

$$3\lambda - 3 + \lambda - 2 + 1 - 2\lambda = 2$$

$$2\lambda = 6$$

$$\lambda = 3$$

$$P(6, 1, -5)$$

$$q = \frac{18 - 4 - 60 - 32}{\sqrt{9 + 16 + 144}} = \frac{78}{13} = 6$$

Equation with roots q and $2q$ is

$$x^2 - 3qx + 2q^2 = 0$$

$$x^2 - 18x + 72 = 0$$

17. If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero vectors and $\hat{n}$ is a unit vector perpendicular to $\vec{c}$ such that $\vec{a} = \alpha\vec{b} - \hat{n}$, ($\alpha \neq 0$) and $\vec{b} \cdot \vec{c} = 12$, then $|\vec{c} \times (\vec{a} \times \vec{b})|$ is equal to :

यदि $\vec{a}, \vec{b}, \vec{c}$ तीन शून्येतर सदिश हैं तथा $\vec{c}$ के लंबवत एक इकाई सदिश $\hat{n}$ है, जिनके लिए $\vec{a} = \alpha\vec{b} - \hat{n}$, ($\alpha \neq 0$) तथा $\vec{b} \cdot \vec{c} = 12$ है, तो $|\vec{c} \times (\vec{a} \times \vec{b})|$ बराबर है :

- (A) 12 (B) 9 (C) 6 (D) 15

Ans. A

Sol. $\vec{n} \cdot \vec{c} = 0, |\vec{n}| = 1$

$$\vec{a} = \alpha\vec{b} - \hat{n} \text{ \& } \vec{b} \cdot \vec{c} = 12$$

$$\vec{a} \cdot \vec{c} = \alpha\vec{b} \cdot \vec{c} - 0$$

$$\vec{a} \cdot \vec{c} = 12\alpha$$

$$|\vec{c} \times (\vec{a} \times \vec{b})|$$

$$|(\vec{c} \cdot \vec{b})\vec{a} - (\vec{c} \cdot \vec{a})\vec{b}|$$

$$|12\vec{a} - 12\alpha\vec{b}|$$

$$12|\vec{a} - \alpha\vec{b}|$$

$$12|\hat{n}| = 12$$

18. A vector $\vec{v}$ in the first octant is inclined to the x-axis at 60° , to the y-axis at 45° and to the z-axis at an acute angle. If a plane passing through the points $(\sqrt{2}, -1, 1)$ and (a, b, c) , is normal to $\vec{v}$, then

प्रथम अष्टांशक में एक सदिश $\vec{v}$ का झुकाव x अक्ष से 60° , y-अक्ष से 45° तथा z-अक्ष से एक न्यून कोण है। यदि बिंदु $(\sqrt{2}, -1, 1)$ व (a, b, c) से होकर जाने वाला समतल $\vec{v}$ पर लंबवत् है, तो

(A) $a + \sqrt{2}b + c = 1$ (B) $\sqrt{2}a + b + c = 1$ (C) $a + b + \sqrt{2}c = 1$ (D) $\sqrt{2}a - b + c = 1$

Ans. A

Sol. $\therefore 1 = \cos 60^\circ = \frac{1}{2}$

Let $\vec{v} = l\hat{i} + m\hat{j} + n\hat{k}$

$m = \cos 45^\circ = \frac{1}{\sqrt{2}}$

$\therefore l^2 + m^2 + n^2 = 1$

$\Rightarrow n^2 = \frac{1}{4}$

$\Rightarrow n = \frac{1}{2}, -\frac{1}{2}$ reject as $\vec{v}$ is in first octant

So, dr's of normal to the plane can be taken as $1, \sqrt{2}, 1$

Hence

$1(a - \sqrt{2}) + \sqrt{2}(b + 1) + 1(c - 1) = 0 \Rightarrow a + \sqrt{2}b + c = 1$

19. If the four points, whose position vectors are $3\hat{i} - 4\hat{j} + 2\hat{k}, \hat{i} + 2\hat{j} - \hat{k}, -2\hat{i} - \hat{j} + 3\hat{k}$ and $5\hat{i} - 2\alpha\hat{j} + 4\hat{k}$ are coplanar, then α is equal to

यदि चार बिंदु, जिनके स्थिति सदिश $3\hat{i} - 4\hat{j} + 2\hat{k}, \hat{i} + 2\hat{j} - \hat{k}, -2\hat{i} - \hat{j} + 3\hat{k}$ तथा $5\hat{i} - 2\alpha\hat{j} + 4\hat{k}$ है, सहतलीय है, तो α बराबर है

(A) $-\frac{73}{17}$ (B) $\frac{107}{17}$ (C) $\frac{73}{17}$ (D) $-\frac{107}{17}$

Ans. C

Sol.
$$\begin{vmatrix} 3-5 & -4+2\alpha & 2-4 \\ 3-1 & -4-2 & 2+1 \\ 3+2 & -4+1 & 2-3 \end{vmatrix} = \begin{vmatrix} -2 & 2\alpha-4 & -2 \\ 2 & -6 & 3 \\ 5 & -3 & -1 \end{vmatrix} = 0$$

$$\begin{vmatrix} 0 & 2\alpha-4 & -2 \\ -1 & -6 & 3 \\ 6 & -3 & -1 \end{vmatrix} = 0 \Rightarrow -(2\alpha-4)(1-18) - 2(3+36) = 0$$

$(\alpha-2)(-17) + 39 = 0$

$\alpha - 2 = \frac{39}{17} \Rightarrow \alpha = 2 + \frac{39}{17} = \frac{34+39}{17}$

$$\alpha = \frac{73}{17}$$

20. If four distinct points with position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are coplanar, then $[\vec{a} \vec{b} \vec{c}]$ is equal to :

यदि चार भिन्न बिंदु, जिनके स्थिति सदिश $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ हैं, सहतलीय हैं, तो $[\vec{a} \vec{b} \vec{c}]$ बराबर है :

(A) $[\vec{a} \vec{d} \vec{b}] + [\vec{d} \vec{c} \vec{a}] + [\vec{d} \vec{b} \vec{c}]$ (B) $[\vec{d} \vec{b} \vec{a}] + [\vec{a} \vec{c} \vec{d}] + [\vec{d} \vec{b} \vec{c}]$

(C) $[\vec{d} \vec{c} \vec{a}] + [\vec{b} \vec{d} \vec{a}] + [\vec{c} \vec{d} \vec{b}]$ (D) $[\vec{b} \vec{c} \vec{d}] + [\vec{d} \vec{a} \vec{c}] + [\vec{d} \vec{b} \vec{a}]$

Ans. C

Sol. $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are coplanar point then

vectors $\vec{b} - \vec{a}, \vec{c} - \vec{a}, \vec{d} - \vec{a}$ are also coplanar

$$[\vec{b} - \vec{a} \quad \vec{c} - \vec{a} \quad \vec{d} - \vec{a}] = 0$$

$$[\vec{b} \vec{c} \vec{d}] - [\vec{b} \vec{c} \vec{a}] - [\vec{b} \vec{a} \vec{d}] - [\vec{a} \vec{c} \vec{d}] = 0$$

$$[\vec{a} \vec{b} \vec{c}] = [\vec{b} \vec{c} \vec{d}] + [\vec{a} \vec{b} \vec{d}] + [\vec{a} \vec{d} \vec{c}]$$